Perfectly Shuffle a Deck of 32 Cards

You may represent each Card through numbers from 1 to 32

$$card_deck = [\dots]$$

A perfect shuffle:

1. splits the card deck in half
2. interleaves all elements from the first and second half

- For example:
  [1,2,3,4] → [1,2] + [3,4]
  [1,2] + [3,4] → [1,3,2,4]

Pseudo-Code

- Create a list containing cards of a regular Skat card deck
- Split the list in half
- Create a list and add the first element of the first half, the second element of the second half etc.

- **Bonus:** What happens if you shuffle the card deck 6 times in a row?

Math background:

Each card in the deck may be presented through numbers from 1 to 32. In one shuffle, the original desk is spited in half and it interleaves all elements from the first and second half into a new sequence.

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main.py assignment 1.txt

```python
# present the deck of cards by numbers
list1 = list (range(1,33))
print("The deck is:", list1)

# split the deck into the first and the second half
mid = len(list1)//2
print("The length of half of the deck is", mid)

# store the two halves into new lists
list2 = list1[:mid]
list3 = list1[mid:]
print("The first half of the original deck is:", list2)
print("The second half of the original deck is:", list3)

# reorgnize the deck method #1
list4 = []
for i in range(len(list2)):
    list4.append(list2[i])
    list4.append(list3[i])
print("The deck after one shuffle is:", list4)

# reorgnize the deck method #2
list5 = [item for pair in zip(list2, list3) for item in pair]
print("The deck after one shuffle is:(method 2)", list5)

# reorgnize the deck method #3
list6 = [item for pair in zip(list1[:16], list1[16:]) for item in pair]
print("The deck after one shuffle is:(method 3)", list6)

# shuffle the deck 6 times
i = list (range(1,7))
print ("i = ", i)

for i in range(1,6): list1 = [item for pair in zip (list1[:16], list1[16:])
    for item in pair];

print(f"The deck after {i} times of shuffle: {list1}")

for i in range(1,7): list1 = [item for pair in zip (list1[:16], list1[16:])
    for item in pair];

print(f"The deck after {i} times of shuffle: {list1}")
```

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```
The deck is: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19,
20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32]
The length of half of the deck is 16
The first half of the original deck is: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13,
14, 15, 16]
The second half of the original deck is: [17, 18, 19, 20, 21, 22, 23, 24, 25, 26,
27, 28, 29, 30, 31, 32]
The deck after one shuffle is: [1, 17, 2, 18, 3, 19, 4, 20, 5, 21, 6, 22, 7, 23, 8,
24, 9, 25, 10, 26, 11, 27, 12, 28, 13, 29, 14, 30, 15, 31, 16, 32]
The deck after one shuffle is:(method 2) [1, 17, 2, 18, 3, 19, 4, 20, 5, 21, 6, 22,
7, 23, 8, 24, 9, 25, 10, 26, 11, 27, 12, 28, 13, 29, 14, 30, 15, 31, 16, 32]
The deck after one shuffle is:(method 3) [1, 17, 2, 18, 3, 19, 4, 20, 5, 21, 6, 22,
7, 23, 8, 24, 9, 25, 10, 26, 11, 27, 12, 28, 13, 29, 14, 30, 15, 31, 16, 32]
i =  [1, 2, 3, 4, 5, 6]
The deck after 5 times of shuffle: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14,
15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32]
The deck after 6 times of shuffle: [1, 17, 2, 18, 3, 19, 4, 20, 5, 21, 6, 22, 7,
23, 8, 24, 9, 25, 10, 26, 11, 27, 12, 28, 13, 29, 14, 30, 15, 31, 16, 32]
```